MODEL QUESTION PAPER 1

S6 INSTRUMENTATION ENGINEERING

SUBJECT: AIRCRAFT INSTRUMENTS

Time: 3 Hour *Max. Marks: 75*

I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)

		Module Outcomes	*Cognitive Level*
1	List out any one type of flight and navigation instrument	M.1.01	R
2	Write any one advantage of straight scale display	M.1.05	R
3	Define static pressure	M.2.02	R
4	Differentiate between a LED and LCD display	M.1.01	U
5	Define term “pitch” in a gyro instrument	M.3.01	R
6	Define critical Mach number	M.2.04	R
7	Define gyroscope	M.3.01	R
8	Define degrees of freedom of a Gyroscope	M.3.01	R
9	List any one principle method for flight data recording	M.4.06	R

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		Module Outcomes	*Cognitive Level*
1.	Write a short note on history of Aircrafts	M1.03	R
2.	Explain LED display used in aircrafts	M1.05	U
3.	List out different type of circular displays	M1.03	R
4.	Explain Pitot pressure	M2.01	U
5.	Explain Tacho Prob	M3.05	U
6.	List out fundamental properties of Gyroscope	M3.01	R
7.	Explain Altitude indicator gyroscope	M3.01	U
8.	List out different methods for driving rotors of gyro flight instrument	M3.02	R
9.	Explain the two types of thermocouples employed in thermoelectric indicating system	M4.02	U
10.	Explain the working principle of an accelerometer used in Aircraft	M4.07	U

Part – C

Answer any six questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		Module Outcomes	*Cognitive Level*
III	Explain colored displays used in Aircrafts	M1.05	U
	OR		
IV	Describe direct displays used in Aircrafts	M1.05	U
V	Elaborate on the sensing and transmission of Pitot & static pressure with suitable diagrams	M2.02	U
	OR		
VI	Explain the operation of a vertical speed indicator under the flight conditions below (a) Level Flight (b) Descent	M2.04	U
VII	Explain the heating circuit arrangement in pitot tube	M2.01	U
	OR		
VIII	Draw and explain Aneroid Barometer	M2.03	U
IX	Explain the fundamental properties of a gyroscope	M3.01	U
	OR		
X	Explain the working of an electrically operated engine speed indicator	M3.04	U
XI	Explain the radiation pyrometer used in exhaust gas temperature measurement	M4.03	U
	OR		
XII	Explain the working of inductor pressure transmitter with a diagram	M4.04	U
XIII	Explain the construction and working a pressure switch	M4.04 M4.05	U
	OR		
XIV	Explain the working of capacitance type fuel gauge system used in		U

	Aircrafts		

MODEL QUESTION PAPER 2

S6 INSTRUMENTATION ENGINEERING

SUBJECT: AIRCRAFT INSTRUMENTS

Time: 3 Hour *Max. Marks: 75*

I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)

		Module Outcomes	***Cognitive Level***
1	Define the term aerodynamics?	M1.04	R
2	List any two flight and navigation instruments?	M1.01	R
3	Define Mach number?	M2.04	R
4	What is the function of Pitot tube in airplanes?	M2.02	R
5	Define degree of freedom?	M3.02	R
6	What does Gyro horizon means?	M3.03	R
7	Define the working principle of thermocouples?.	M4.02	R
8	Define the principle of Gyroscope?	M3.01	R
9	Write the principle of operation of radiation pyrometer?	M4.03	R

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		Module Outcomes	***Cognitive Level***
1.	Explain quantitative displays?	M1.05	U
2.	Define the three basic forces: thrust, drag, lift, considered in aerodynamics?	M1.04	R
3.	Explain Pitot pressure?	M2.01	U
4.	How an Aneroid barometer works?	M2.03	R
5.	List the properties of Gyroscope?	M3.02	R
6.	Define Pitch, Bank and Turn?	M3.01	R
7.	Explain the importance of flight data recording?	M4.06	U
8.	List the different types of temperature sensors used in air crafts?	M4.01	R

9.	Explain the working principle of accelerometer used in Aircraft?	M4.07	U
10.	List the different types of thermocouples employed in thermoelectric indicating system?	M4.02	R

Part – C

Answer any six questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		Module Outcomes	*Cognitive Level*
III	Explain LCD and LED displays used in Aircrafts	M1.05	U
	OR		
IV	Illustrate Head up display in an aircraft?	M1.05	U
V	Explain construction of Pitot static Probe used in air craft instruments?	M2.02	U
	OR		
VI	Explain the working of Mach meter?	M2.04	U
VII	Explain Tacho probe and Indicator System?.	M3.05	U
	OR		
VIII	Explain degrees of freedom of Gyroscope?	M3.01	U
IX	Explain working of Radiation Pyrometer for exhaust gas temperature measurement?	M4.03	U
	OR		
X	Explain the working of Pressure switches?	M4.04	U
XI	Explain the pneumatic method of driving Gyroscope?.	M3.02	U
	OR		
XII	Explain working of electrically operated Engine speed indicators in aircrafts?	M3.04	U
XIII	Explain the working of capacitance type fuel gauge system used in Aircrafts?	M4.05	U
	OR		
XIV	Explain in detail the working principle of pressure transmitter	M4.04	U

Question Wise Analysis

Course: **AIRCRAFT INSTRUMENTS** – Question - 2

Q. No	**Module Outcome**	**Cognitive Level**	**Marks**	**Time(Min.)**
I.1	M1.04	R	1	2
I.2	M1.01	R	1	2
I.3	M2.04	R	1	2
I.4	M2.02	R	1	2
I.5	M3.02	R	1	2
I.6	M3.03	R	1	2
I.7	M4.02	R	1	2
I.8	M3.01	R	1	2
I.9	M4.03	R	1	2
II.1	M1.05	U	3	8
II.2	M1.04	R	3	8
II.3	M2.01	U	3	8
II.4	M2.03	R	3	8
II.5	M3.02	R	3	8
II.6	M3.01	R	3	8
II.7	M4.06	U	3	8
II.8	M4.01	R	3	8
II.9	M4.07	U	3	8
II.10	M4.02	R	3	8
III	M1.05	U	7	16
IV	M1.05	U	7	16
V	M2.02	U	7	16
VI	M2.04	U	7	16
VII	M3.05	U	7	16
VIII	M3.01	U	7	16

IX	M4.03	U	7	16
X	M4.04	U	7	16
XI	M3.02	U	7	16
XII	M3.04	U	7	16
XIII	M4.05	U	7	16
XIV	M4.04	U	7	16

Course : **AIRCRAFT INSTRUMENTS – QP 2.**

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Mark Distribution

Module	hr / Module	Marks/Module $(h_i / \sum H_i) * 123$ (±5%)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	17	28 ± 6	3	3	3	9	2	14	8	26
2	19	31 ± 6	2	2	1	3	4	28	7	33
3	20	33 ± 6	3	3	4	12	2	14	9	29
4	19	31 ± 6	1	1	2	6	4	28	7	35
Total	75	123	9	9	10	30	12	84	31	123

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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	27	22
Understanding	96	78
Applying	0	0
Analysing	0	0
Evaluating	0	0
Creating	0	0
Total	123	100